



Fuzzy Logic & Neural Networks (CS-514)

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Introduction

What is “Fuzzy” and “Fuzzy Logic”

- ▶ **“Fuzzy” means not clear or not exact.**
- Imagine you are dealing with the concept of "tall" when referring to the height of people.
- ▶ **In normal logic, things are either yes or no, true or false, 0 or 1. But in fuzzy logic, things can be in between.**
- In classical logic, you might define a strict cutoff where someone taller than, say, 6 feet is considered "tall" (true), and anyone shorter is "not tall" (false).
- **So, fuzzy = dealing with maybes, shades, and in-betweens instead of strict yes/no.**

Introduction

Fuzzy Logic (Historical Developments)

Year	Person / Group	Development
1965	Lotfi A. Zadeh (UC Berkeley)	Introduced the concept of Fuzzy Sets in his famous paper.
1973	Lotfi A. Zadeh	Proposed Fuzzy Logic as an extension of classical logic.
1975	Ebrahim Mamdani (UK)	Built the first fuzzy control system for a steam engine.
1980s	Japanese Companies (Hitachi, Matsushita, etc.)	Applied fuzzy logic in industrial control systems , elevators, trains, washing machines.
1990s	Worldwide Adoption	Fuzzy logic used in consumer electronics, automotive systems, and expert systems .
2000s– Now	Researchers Globally	Combined fuzzy logic with AI, neural networks, and optimization for smarter decision-making.

Introduction

Fuzzy Logic

- ▶ **In Fuzzy logic value can lie anywhere between 0 and 1.**
- In fuzzy logic, the concept of "tall" can be considered as:
- A person who is 5'10" might be considered 0.6 tall (partially tall).
- A person who is 6'2" might be considered 0.9 tall (almost fully tall).
- A person who is 6'5" might be considered 1.0 tall (fully tall).

Introduction

Fuzzy Logic

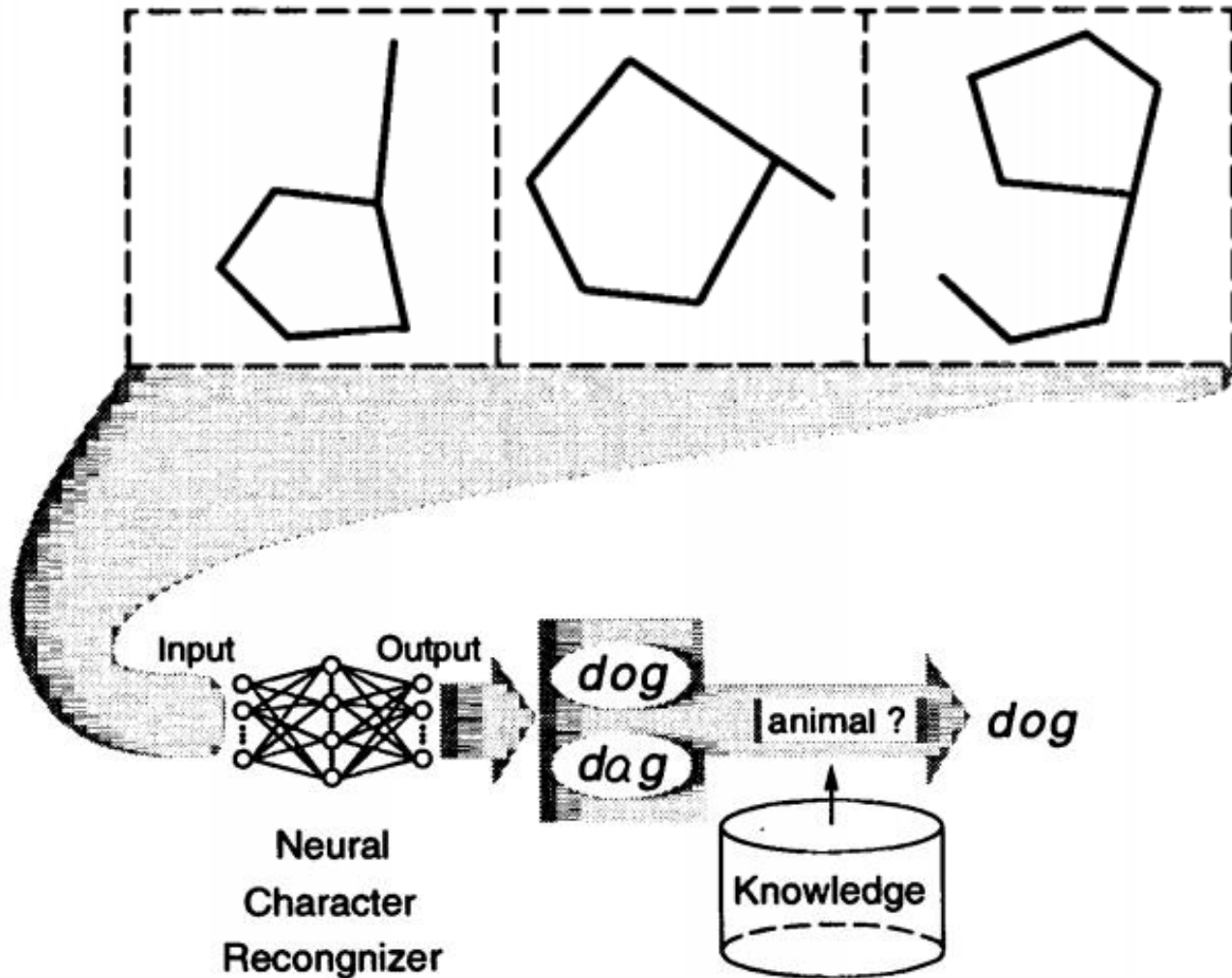
- ▶ **Is tolerant of imprecision, uncertainty, partial truth, and approximation.**
- Here, the ambiguity comes from the fact that the term "tall" **doesn't have a clear boundary**—it varies in degrees, capturing the fuzzy nature of real-world categorizations.

Benefits:

- ▶ To model the imprecision inherent in many real-world scenarios.
- ▶ To implement human reasoning in machines.

Introduction

Why Fuzzy?



Introduction

Fuzzy Logic Applications:

- **Autonomous Driving:** Fuzzy logic is used in decision-making processes for autonomous vehicles. This includes making real-time decisions about speed, braking, lane changes, and obstacle avoidance.

Companies like **Tesla** and **Waymo** are integrating fuzzy logic principles as part of their AI systems for handling uncertain situations, such as varying weather conditions or unpredictable human behavior on the road.

Introduction

Fuzzy Logic Applications:

- **Washing Machines:** Fuzzy logic is used to determine the optimal wash cycle based on the amount of dirt, type of fabric, and load.
- **Air Conditioning:** The system can regulate temperature more smoothly, considering factors like room temperature, humidity, and even human presence.
- **Cameras:** In autofocus systems for digital cameras, fuzzy logic helps to improve focusing by considering various degrees of sharpness and lighting conditions.

Recap of Fundamentals

Set: A collection of objects in a given domain with similar qualities.

Features:

- Crisp boundaries.
- An element either belongs to the set or it does not.

Universe of Discourse: The set/range of all possible objects/values having the same characteristics on a given problem/condition.

Examples:

- Positive integer numbers.
- Temperature of this classroom.

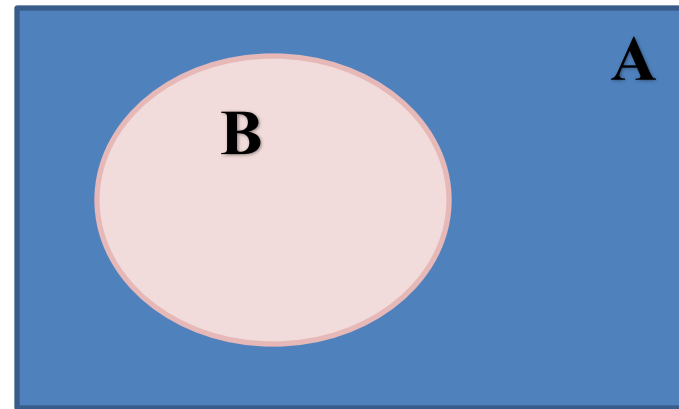
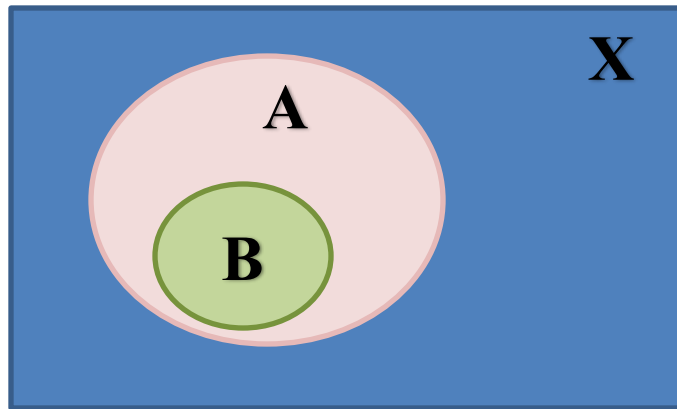
Recap of Fundamentals

- The individual elements in the universe X will be denoted as x .
- The values/features of the elements in X can be discrete, countable integers or continuous valued quantities on the real line.
- The total number of elements in a universe X is called its cardinal number.

Recap of Fundamentals

Examples:

- Students of this Institute. (X)
- Students of this class. (A)
- Female students of this class. (B)



$A = \{\text{Mahesh, Ramesh, Suresh, Sita, Gita, Rita, Meeta}\}$

$B = \{a : \text{gender}(a) = \text{female}\}$

So, cardinality of A is 7 and B is 4.

Recap of Fundamentals

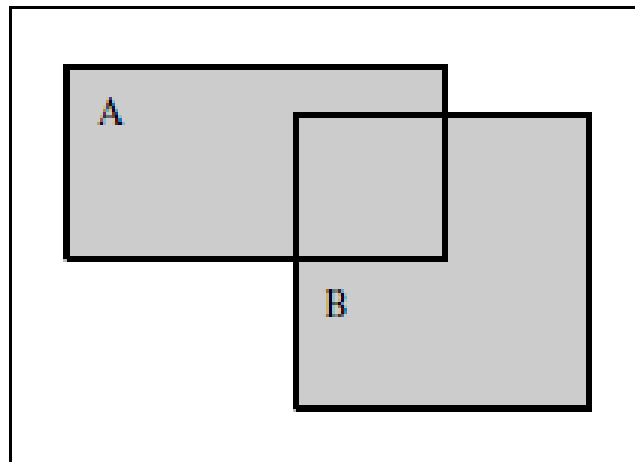
Operation on Classical Sets

Union

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

The union between the two sets, denoted $A \cup B$, represents all those elements in the universe that reside in (or belong to) the set A, the set B, or both sets A and B.

This operation is also called the logical **OR**



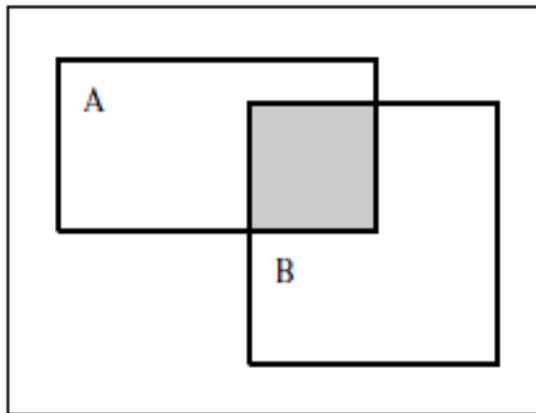
Recap of Fundamentals

Operation on Classical Sets

Intersection $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

The intersection of the two sets, denoted $A \cap B$, represents all those elements in the universe X that simultaneously reside in (or belong to) both sets A and B .

This operation is also called the logical **AND**



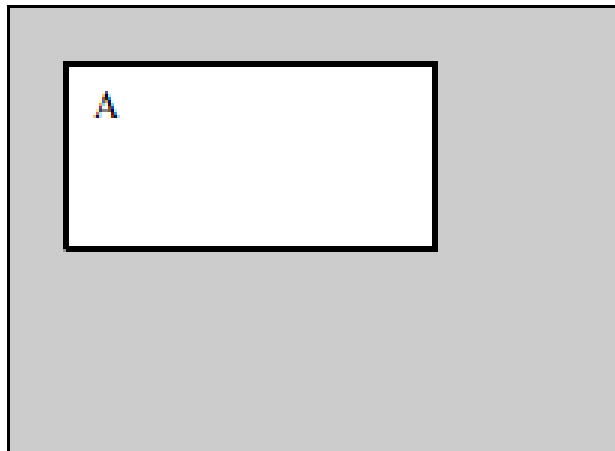
Recap of Fundamentals

Operation on Classical Sets

Complement

$$\bar{A} = \{x \mid x \notin A, x \in X\}$$

The complement of a set A , is defined as the collection of all elements in the universe that do not reside in the set A .



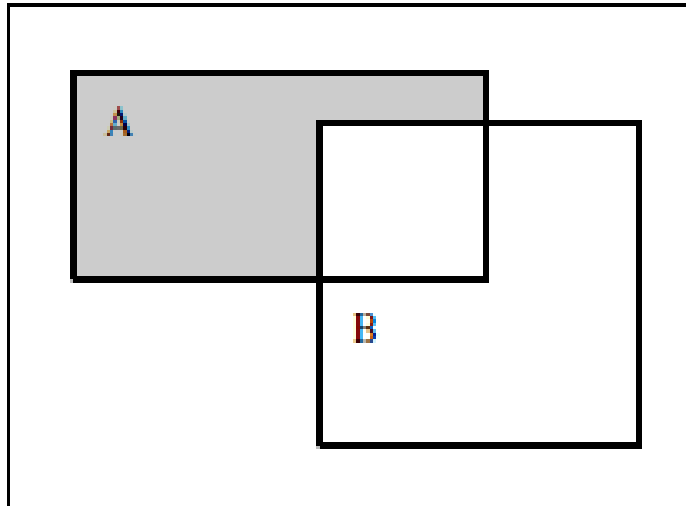
Recap of Fundamentals

Operation on Classical Sets

Difference

$$A \mid B = \{x \mid x \in A \text{ and } x \notin B\}$$

The difference of a set A with respect to B, denoted as $A \mid B$, is defined as the collection of all elements in the universe that reside in A and that do not reside in B simultaneously.



Recap of Fundamentals

Properties of Classical Sets

Commutativity $A \cup B = B \cup A$

$$A \cap B = B \cap A$$

Associativity $A \cup (B \cup C) = (A \cup B) \cup C$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributivity $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Idempotency $A \cup A = A$

$$A \cap A = A$$

Identity $A \cup \emptyset = A$

$$A \cap X = A$$

$$A \cap \emptyset = \emptyset$$

$$A \cup X = X$$

Recap of Fundamentals

Properties of Classical Sets

Transitivity If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$

Law of Excluded Middle

$$A \cup \bar{A} = X$$

Law of Contradiction

$$A \cap \bar{A} = \emptyset$$

DE Morgan's Laws

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B}$$

$$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

Fuzzy Sets

From Classical Sets to Fuzzy Sets:

Let X be the universe of discourse and x represents its element such that $x \in X$

Then a classical set A can be defined as:

$$A = \{x \mid x \text{ meets certain condition and } x \in X\}$$

For example, if we consider a set A of positive integers below 20 and greater than 15.

Therefore, the classical set A can be defined as:

$$A = \{x \mid 15 < x < 20 \text{ and } x \in X\} \quad A = \{16, 17, 18, 19\}$$

Here universe of discourse X is the set of all positive integers

Fuzzy Sets

From Classical Sets to Fuzzy Sets:

For the given set $A = \{ 16, 17, 18, 19 \}$

We can say that the membership value of an integer belonging to set A is one (True) and for all other integers the membership value is zero (False).

Since Fuzzy theory is based on Multi-valued logic. Hence, we can define a Fuzzy set A of positive integers below 20 and greater than 15, where every element is either completely present or partially present as:

$$A = \{ (16, 0.7), (17, 0.1), (18, 0.2), (19, 0.9) \mid x \in X \}$$

Fuzzy Sets

Definition:

A collection of objects with a smooth boundary.
If X is a collection of objects denoted by x , then a fuzzy set A in X is defined as a set of ordered pairs:

$$A = \{(x, \mu_A(x)) \mid x \in X\}$$

Let $X = \{\text{Mumbai, Ahmadabad, Surat, IIIT Surat}\}$
be the UOD for places one may choose to live.

Define a Fuzzy set A = “Desirable place to live”

Fuzzy Sets

Let $X = \{\text{Mumbai, Ahmadabad, Surat, IIIT Surat}\}$ be the UOD for places one may choose to live.

Define a Fuzzy set $A = \text{“Desirable place to live”}$

$A = \{(\text{Mumbai}, 0.9), (\text{Ahmadabad}, 0.7), (\text{Surat}, 0.5), (\text{IIIT Surat}, \underline{\quad})\}$